

packages/graph-view/src/utils/louvain.ts

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Overview

A new utility `packages/graph-view/src/utils/louvain.ts` introduces Louvain community detection for a subset of graph nodes and edges. It exports `detectCommunities`, which builds an undirected graph, runs `graphology-communities-louvain`, and returns a `Map<string, number>` of node-to-community assignments.

Key changes

- **New imports:** `Graph` from `graphology`, `louvain` from `graphology-communities-louvain`, and `GraphEdge` type from `@novadiff/graph-core/types`.
- **`detectCommunities` implementation:**
 - Builds a graph with `new Graph({ type: "undirected", multi: false })`.
 - Adds only nodes in `nodeIds` and edges whose endpoints are both in that set, filtering out self-loops and duplicates.
 - Calls `louvain(g)` and maps the result to a `Map`.
 - Defensive reassignment: any `-1` sentinels are replaced with unique IDs beyond the current maximum.
- **Export:** The function is exported for external use.

Impact

- **Correctness:** Guarantees unique community IDs even if the underlying library omits nodes or returns `-1`.
- **Performance:** Graph construction and edge filtering add $O(|E|)$ overhead; acceptable for moderate graph sizes but may impact large datasets.
- **Maintainability:** Centralizes community detection logic; future updates to `graphology-communities-louvain` can be isolated here.
- **Compatibility:** No changes to existing APIs; simply adds a new helper.
- **Observability:** No logging; consumers should handle the returned `Map` directly.

Risks & follow-ups

- **Library version drift:** Verify that `graphology-communities-louvain` v2 still returns `Record<string, number>` and that the defensive logic remains necessary.
- **Edge cases:** Test with disconnected nodes, self-loops, and duplicate edges to ensure the filtering logic behaves as intended.
- **Performance regression:** Benchmark on large node sets to confirm the graph construction cost is acceptable.

- **Type safety:** Ensure `GraphEdge` type aligns with the edges passed to `detectCommunities`; mismatches could cause runtime errors.